

FIRST SOLVE

New here? Do the walkthrough once at [cubepath-s3.vercel.app/learn](#). This could save the memory jog.

NOTATION
R U F L D B = turn that face clockwise, looking at it from outside the cube.
= counter-clockwise, **2** = half turn. Lowercase **r f l d b** = that face plus the layer behind it, turning together
y' = spin the whole cube left, white staying down. A whole-cube turn isn't nothing.

WORDS
 algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. NO algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on TOP turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'LUL



white sticker UP run the first one once to knock it out, then look again
RUR'U'



white sticker DOWN run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center below it. The arrow shows which slot it goes to.



goes RIGHT
U' R'IGHT y L'U'LU



goes LEFT
U' L'U'LU y' RUR'U'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

RUR'U' sexy move **RUR'U** sane **R'RFR'** sledgehammer gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f'

Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\otimes$ 1 yellow, rest clockwise
RUR'U R'U'2R'

Anti-Sune $\otimes$ 1 yellow, rest counter-clockwise
RU2R' U'RUR'



Pi $\otimes$ 0 yellow, headlights left
fRUR'U'f' fRUR'U'f'

OLL CORNERS - continued



Headlights $\otimes$ 2 yellow, headlights at the back
R2DR'U2 R'D'R'U2R'



2x Headlights $\otimes$ 0 yellow, headlights both sides
RUR'U RUR'R URU2R'



Chameleon $\otimes$ 2 yellow next to each other
rUR'U'r' FRF'



Bowtie $\otimes$ 2 yellow diagonal
f'rUR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.

Turn not yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

fRUR'U **sexy move** **RUR'U Sune** **f'FRF'** **algorithm** **can = chamele** **grin**

ADJACENT CORNER SWAP

exactly one face shows headlight

• T 1. headlight; pairs R-left F-left
RUR'U' R'F R2UR'U' RUR'F'

• Ja 1. solid; pair R-right F-left R-front
x R2FR'F' RU2 r'Ur U2

• Jb 1. solid; pairs R-left F-right R-back
RUR'F' RUR'U' R'F R2UR'

• Aa 1. headlight; pairs F-right R-front
x L2D2 L'U'L D2 L'U'L'

• Ab 1. solid; pairs R-right R-back
x' L2D2 LUL' D2 LU'L

• F1. solid
R'U'F' RUR'U' R'F R2UR'U'

One face shows headlight; two corners of that face match. Hold as drawn, run, then turn the top to finish.

■ **RUR'U'** sexy move
 ■ **RUR'U' Sune**
 ■ **R'FR'F** sledehammer
 eap = change rise

ADJACENT CORNER SWAP

continued

• Ra 1. headlight; pairs R-left
R'U'U'R' RUR DR'UR'D' R'U2R'

• Rb 1. headlight; pairs R-left
R2F RUR'U' FR U2R'U2R

• Ga 1. headlight; pairs R-right
R2UR'U'U' RUR'2 U'D R'UR'D'

• Gb 1. headlight; pairs R-back
R'U'R UD' R2UR'URU' R'UR'2D

• Gc 1. headlight; pairs R-right
R2U'RURU' RUR'2 UD' R'UR'D

• Gd 1. headlight; pairs R-front
RUR' U'D R2UR'URU' UR'UR'2D

Learns in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ANNEX – BIG CUBES AND THE MAP
GOO A A

BIG CUBES, 4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen it.

last 2 edges

$$\begin{matrix} R'w & U & R' & U & R' & F & R' & F \\ 2R2 & U2 & 2R2 & Uw2 & 2R2 & Uw2 \end{matrix}$$

slice out, run it, slice back – both cubes

2 edge pairs swapped- half the time

4x4 PLL parity

4x4 OLL parity = 1 edge pair flipped- half the time

$$\begin{matrix} Rw & U2 & x & Rw & U2 & Rw & U2 & Rw \\ 5R5 & edge & parity & 1 & edge & pair & flipped & - half & the & time \end{matrix}$$

one *edge* differs, 3x5 has no PLL parity

$$\begin{matrix} Rw & U2 & x & Rw & U2 & 3Rw & U2 & Lw &end{matrix}$$

2 edge pairs swapped- half the time

Fix parity during the last layer, before OLL, PLL, nothing to algorithm move corners

NOTATION



Each letter, D and that face clockwise, see from outside the cube – so from your seat L, L and B look counter-clockwise = counter-clockwise, z = half turn.

M = middle slice turning the way I turns, x y z = the whole cube, turning the way R, U and F. Lowercase i f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT – use it until you know Suno

Twist a yellow corner four turns at a time, turning ONLY the top face from corners.

yellow faces RIGHT

yellow faces FRONT

D'R'D x2

D'R'DR x2

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger - 3x3x3 = 3x3x3

RUR'U' sexy move, **RUR'U'** sexy, **R'F'F'** slice/shrinker, **sexy** = change spin

